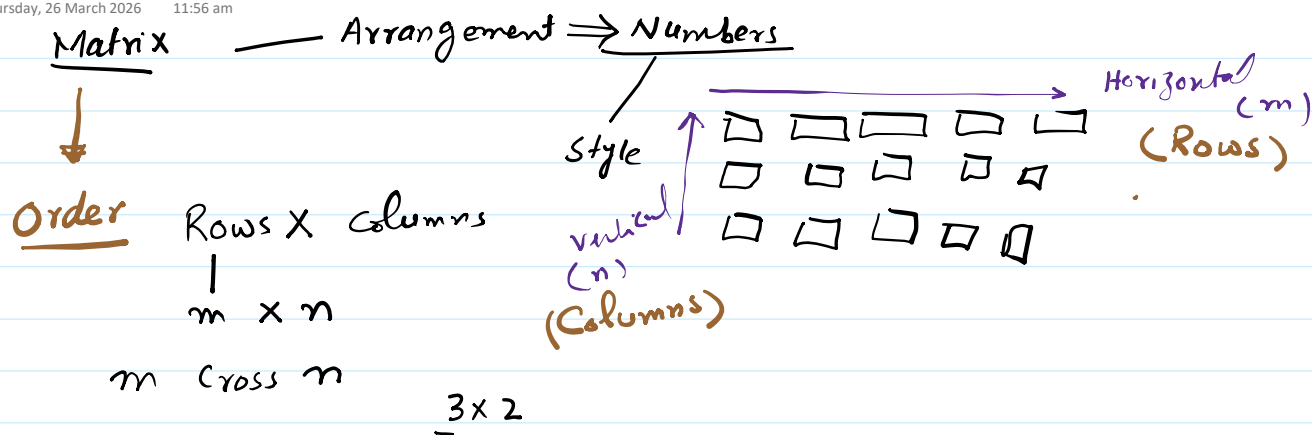




① Square Matrix $m = n$

a) Zero matrix O_1 $\begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$ O_2 $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 2}$ O_3 $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

b) Identity matrix I_1 $\begin{bmatrix} 1 \end{bmatrix}_{1 \times 1}$ I_2 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$ I_3 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$

② Rectangular matrix $m \neq n$

Special matrices

a) Row matrix : order $1 \times n$ $n > 1$

$$[a_{1j}] \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \end{bmatrix}$$

Example $\begin{bmatrix} 5 & -2 & 3 & 4 \end{bmatrix}$

b) Column matrix order $m \times 1$ $m \geq 2$ $[a_{i1}] \quad i \geq 1$

$$\therefore \begin{pmatrix} 5 \\ -2 \\ 1 \\ 3 \end{pmatrix}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix}$$

$$\begin{pmatrix} 3 & -1 & 2 & 4 \\ 1 & 5 & 6 & 2 \end{pmatrix}$$

order = 2×4

(Equal matrices)

- ① order same
- ② corresponding element same

Operations on matrices

① Transpose of a matrix A matrix of order $m \times n$

Then $A^t = n \times m$

$$A = \begin{pmatrix} 4 & 5 \\ 2 & 1 \end{pmatrix}$$

$$A^t = \begin{pmatrix} 4 & 2 \\ 5 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \quad A^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 2 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & 1 \end{pmatrix}; A^t = \begin{pmatrix} -1 & 1 & 3 \\ 2 & -1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

② Scalar multiplication

$$K \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \begin{pmatrix} Ka & Kb \\ Kc & Kd \end{pmatrix}$$

Scalar matrix

$$\begin{pmatrix} 2 & 6 \\ -4 & -8 \end{pmatrix} = 2 \begin{pmatrix} 1 & 3 \\ -2 & -4 \end{pmatrix}$$

scalar matrix

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \Rightarrow 2 \begin{pmatrix} 1 & 3/2 \\ 2 & 3 \end{pmatrix}$$

Not a scalar matrix

③ Addition & Subtraction

Condition When order of all matrices is same.

How to add / Subtract
+, -

Corresponding Element

(whose address is same)

$$\begin{pmatrix} -2 & 3 & -1 \\ 1 & 2 & 5 \\ -3 & 1 & 2 \end{pmatrix} + \begin{pmatrix} -3 & 4 & -5 \\ -2 & -3 & 7 \\ -2 & 1 & -3 \end{pmatrix}$$

Row no & Column no

$$= \begin{pmatrix} -5 & 7 & -6 \\ -1 & -1 & 12 \\ -5 & 2 & -1 \end{pmatrix}$$

Q. If $A + B = O$; The A and B are additive inverses

$5 + (-5) = 0$

↑ ↑

12 -3 \ 0 | -2 3 \

which gives us
Additive Identity
 $5 + 0 = 0 + 5 = 5$

$$A = \begin{pmatrix} 2 & -3 \\ -1 & 4 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 3 \\ 1 & -4 \end{pmatrix}$$

Additive Identity
 $5 + 0 = 0 + 5 = 5$

Example

$$A = \begin{pmatrix} -2 & 1 \\ x & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & y \\ 3 & 5 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 5 \\ 6 & 2t-1 \end{pmatrix}$$

If $A + B = C$ find x, y, z, t

$$\begin{pmatrix} -1 & 1+y \\ x+3 & 7 \end{pmatrix} = \begin{pmatrix} z & 5 \\ 6 & 2t-1 \end{pmatrix}$$

$$\begin{cases} z = -1 \\ 1+y = 5 \end{cases} \Rightarrow y = 4$$

$$\begin{cases} x+3 = 6 \\ 7 = 2t-1 \end{cases} \Rightarrow \begin{cases} x = 3 \\ 8 = 2t \\ 4 = t \end{cases}$$

Multiplication of Matrices

Condition $A \Rightarrow m_1 \times n_1$

$B \Rightarrow m_2 \times n_2$

CR $\Rightarrow$ Checkin

AB is possible if $n_1 = m_2$, Then Result will be $m_1 \times n_2$

How to multiple RC to multiply

Multiply Corresponding Elements of 1 Row with 1 Column and add them

$$(FF + SS + TT)$$

$$A = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2+2 & -2+1 \\ 3+4 & -3+2 \end{pmatrix}$$

AB is possible $\begin{pmatrix} 2 \times 2 \\ 2 \times 2 \end{pmatrix} = 2 \times 2$

$\begin{bmatrix} R_1 C_1 \\ R_2 C_1 \end{bmatrix} \quad \begin{bmatrix} R_1 C_2 \\ R_2 C_2 \end{bmatrix}$

$$BA = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix}$$

Muneab
 Gull Fatima
 Arham

$$AB = \begin{pmatrix} 0 & 3 \\ 7 & -1 \end{pmatrix} \quad \left| \quad \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix} \right| \quad \begin{matrix} \text{Gull Fatima} \\ \text{Arham} \end{matrix}$$

$$= \begin{pmatrix} -2-3 & 1-2 \\ -4+3 & 2+2 \end{pmatrix}$$

$$= \begin{pmatrix} -5 & -1 \\ -1 & 4 \end{pmatrix}$$

$$AB \neq BA$$

matrices don't obey commutative law of multiplication

① $A = \begin{pmatrix} -1 & 2 \\ 1 & 3 \end{pmatrix}$ find $A^2 = \begin{pmatrix} 3 & 4 \\ 2 & 11 \end{pmatrix}$

$$A^2 = \begin{pmatrix} 1 & 4 \\ 1 & 9 \end{pmatrix}$$

AA

② $1 \times (N) = N$ "1" multiplicative identity

$$5 \times \left(\frac{1}{5}\right) = 1$$

multiplicative inverses

if A & B are 2×2 matrices
The I of 2×2

$$AB = I$$

$$\begin{pmatrix} 3 & 4 \\ -3 & -5 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

B is inverse of A

$$\begin{pmatrix} 3a+4c & 3b+4d \\ -3a-5c & -3b-5d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} 3a+4c=1 \\ -3a-5c=0 \end{cases} \quad \begin{cases} 3b+4d=0 \\ -3b-5d=1 \end{cases}$$

$$-c=1 \quad \boxed{c=-1}$$

$$3a-4=1 \quad 3a=5 \quad a=\frac{5}{3}$$

$$-d=1 \quad \boxed{d=-1}$$

$$3b-5=0 \quad 3b=5 \quad b=\frac{5}{3}$$

$$\begin{pmatrix} \frac{5}{3} & -1 \\ \frac{5}{3} & -1 \end{pmatrix}$$

Division of matrices is not possible! why?

$$3a = 5 \quad a = \frac{5}{3}$$

$$\begin{pmatrix} 3 & 4 \\ -3 & -5 \end{pmatrix} \begin{pmatrix} \frac{5}{3} & \frac{4}{3} \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 5-4 & 4-4 \\ -5+5 & -4+5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$$

$$A^2 - B^2 = (A+B)(A-B)$$

$$= A^2 - AB + BA - B^2$$

$$AB \neq BA \quad \text{so, } A^2 - B^2 \neq (A+B)(A-B)$$

How to find determinant by finding
Co-factor of a matrix

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